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Practical Computational Methods for Social Scientists

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Analysis

Outline



1. Descriptive inference:
 1. Word clouds
 2. Simple summaries
 3. (Dis)similarity measures
 4. Discriminating words
2. Supervised techniques:
 1. Sentiment analysis
3. Unsupervised techniques
 1. Topic modelling

There are so many other techniques and methods we could use. Text analysis is a huge topic. We will add to these materials over time but please suggest ideas for other approaches you would like to learn.

Descriptive inference



Descriptive inference in text analysis refers to the process of summarizing and identifying patterns, structures, and key characteristics in textual data without making causal claims (Grimmer & Stewart, 2013).

Simple summaries include word frequencies, discriminating words etc.

To compute these summaries we use linear algebra on the DTM / DFM.

One of the simplest summaries of the DTM / DFM is a word cloud. This is a visualisation of the frequency counts of a bag of words representation of a corpus.

victims	viet	vietnam	village	virginia
1	2	2	1	



What are the advantages and disadvantages of a word cloud as a means of text analysis?

Descriptive inference: simple summaries



term1	term2	term3	term4	term5
3	0	1	4	0
3	3	2	1	4
0	4	0	2	0
2	1	2	2	3
0	3	2	1	3

Let's say we have a simple DTM / DFM with five documents and five terms. Answer the following questions:

- How many times is term 3 mentioned in the corpus?
- How many terms and tokens are in document 5?
- Compute the row and column totals and write down their interpretations.

As you can see we can use some simple algebra to calculate some quantities or summaries of interest.

Descriptive inference: (dis)similarity



A useful similarity metric in the vector space model is cosine similarity.

$$\cos(W_1, W_2) = \frac{W_1 * W_2}{||W_1|| * ||W_2||}$$

It ranges between 0 (completely different) and 1 (completely similar).

Cosine distance = 1 – cosine similarity

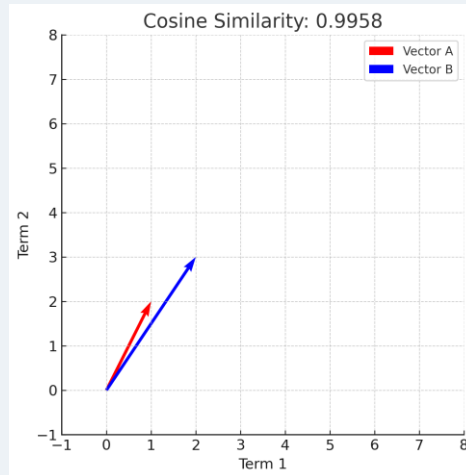
How different (or similar) are documents?

Where W_1 is the vector of document 1 and $||W_1||$ is the normalised length of vector W_1 . It is normalised because we want to compare documents (and therefore vectors) of different lengths.

There are lots of difference measures / metrics of similarity or distance:

- Euclidean
- Manhattan
- Jaccard

Descriptive inference: (dis)similarity



If vectors are sequences of terms, we can compute the angle between these sequences as a measure of how similar they are.

Descriptive inference: discriminating words



Discriminating words = words that characterise the language use in a group of documents in the corpus (Grimmer et al., 2022).

Words that are more prevalent in certain types of documents than others
e.g., do large charities describe their overseas activities differently to medium or small organisations?

We want to explain / predict documents belonging to certain categories, rather than discover these categories.

Descriptive inference: discriminating words



Mutual Information (MI) measures how much information the presence of a word provides about the category (or document) it appears in. High MI scores indicate words that are strongly associated with one category over others.

Fightin' Words find words that are overrepresented in one document compared to another. It is particularly useful for analysing differences in word usage between two documents (Monroe et al., 2008).

What is a limitation of MI? A downside of MI is it only considers the presence or absence of words, not occurrence (and therefore not the probability of the word occurring).

Fightin' Words = Feature Weighting using Log-Odds Ratio with Informative Dirichlet Priors

Descriptive inference: discriminating words



Considerations when using discriminating words approaches:

- Rare words: represent genuine differences in language / discourse or just random chance?
- There is a difference in words that distinguish between categories and those that are indicative of categories.

Supervised techniques: sentiment analysis



We can think of documents as belonging to categories or classes (Spirling, 2022). We can know categories or classes and measure the extent to which our documents align with these.

For example, is a product review positive or negative?

To answer we can look at words used in the review and see how they align with a dictionary of words or statements **that are known to be positive or negative.**

Because the categories and classes are known we use supervised techniques.

Essentially a large corpus of words or statements have been labelled as positive or negative and then we compare the words / statements in our documents to these to produce a sentiment score.

When I say “we” I mean a machine learning algorithm or classifier.

Supervised techniques: sentiment analysis



$$Y = X + e$$

Where:

Y = Labelled outcome e.g., 1 = positive review

X = Document features e.g., words and/or frequencies

e = Uncertainty in our prediction

At its simplest we want to model the relationship between inputs and an output (or covariates and an outcome).

Supervised techniques: sentiment analysis



Basic idea: use a set of pre-defined words with specific connotations to classify our documents automatically, quickly and accurately (Spirling, 2022).

Dictionary-based method.

Aim: measure *whether* and *the extent* to which a document is associated with a given category (sentiment).

$$\sum_{m=1}^M \frac{S_m W_{im}}{N_i}$$

At its simplest we want to model the relationship between inputs and an output (or covariates and an outcome).

For document i and words $m = 1, \dots, M$ in the dictionary,

S_m is the sentiment score of word m .

N_i = Total number words in document that are in dictionary.

Q: N_i is a normalising parameter – why do we need this?

Sentiment scores usually range from -1 (very negative) to +1 (very positive).

Let's look at an example.

Supervised techniques: sentiment analysis



“The Pogues’ Body of an American is a raw, energetic anthem. The driving rhythm, powerful lyrics, and Shane MacGowan’s passionate vocals make it an unforgettable and emotional listening experience.”

- Is this a positive review?
- What words are you using in particular to make your judgement?
- Do you notice any downsides of treating text in this manner?

Some issues (in this text and in general):

- Loss of context (is unforgettable a positive or negative word on its own?).
- Sarcasm?
- Granularity i.e., can this approach really distinguish between categories close together?
- How do we know what sentiment scores to assign?
- How do we know which dictionary is best to use?

Supervised techniques: sentiment analysis



"The Pogues' Body of an American is a raw, energetic anthem. The driving rhythm, powerful lyrics, and Shane MacGowan's passionate vocals make it an unforgettable and emotional listening experience."

Overall score = 0.24

Word scores = {"raw": -0.1, "energetic": 0.8, "anthem": 0.5, "driving": 0.3, "rhythm": 0.2, "powerful": 0.9, "lyrics": 0.3, "passionate": 0.8, "vocals": 0.4, "unforgettable": 1.0, "emotional": 0.7, "listening": 0.2, "experience": 0.3}

How would you interpret the overall score?

Are you in agreement with the individual word scores?

Supervised techniques: sentiment analysis



Some issues (for this example and in general):

- Loss of context (is unforgettable a positive or negative word on its own?).
- Detect sarcasm?
- Granularity i.e., can this approach really distinguish between categories close together?
- How do we know what sentiment scores to assign?
- How do we know which dictionary is best to use?

Sentiment analysis is a simple, somewhat crude approach to classifying text but can be useful in certain circumstances.

We'll revisit sentiment analysis in the practical.

Supervised techniques: topic modelling



Topic modelling is an unsupervised machine learning technique used in text analysis to automatically identify hidden themes or topics within a collection of documents (Grimmer et al., 2022).

It analyses word co-occurrence patterns and grouping words that frequently appear together, forming coherent topics.

Assumes that each document is a mixture of topics and each topic is a mixture of words (Chang et al., 2009).

Therefore documents can belong to multiple topics (or classes) – in contrast to cluster analysis which assigns documents to a single class or group. You often hear topic modelling referred to as a “mixed membership” model.

Supervised techniques: topic modelling



	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
0	support	support	support	support	fund
1	overseas	education	people	provide	school
2	education	provide	new	financial	education
3	work	fund	education	work	training
4	fund	medical	training	education	local
5	people	health	program	international	community
6	new	care	fund	local	help
7	also	program	overseas	community	provide
8	local	also	research	program	providing
9	training	community	providing	people	work

Supervised techniques: topic modelling



Strengths:

- Process large, unstructured text data without requiring prior labelling.
- Very useful for reducing high-dimensional text.
- Reveal meaningful patterns that humans may not be able to detect (certainly at scale).

We can characterise a document well even if we have never seen another like it before (it's an inductive, data-driven approach).

Supervised techniques: topic modelling



Limitations:

- Topics may not be interpretable.
- Inference is tricky.
- Results are sensitive to how many topics you want to identify.
- Probabilistic approach to word generation.

Supervised techniques: topic modelling



Validation:

1. Read through the vocabularies (unique terms) associated with each topic.
2. Read representative documents:
 1. Probabilistic sampling
 2. Documents with the highest proportions in each topic
3. Label each topic with an appropriate name.

Conclusion:

<https://github.com/SGSSOnline/text-analysis/tree/main>

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